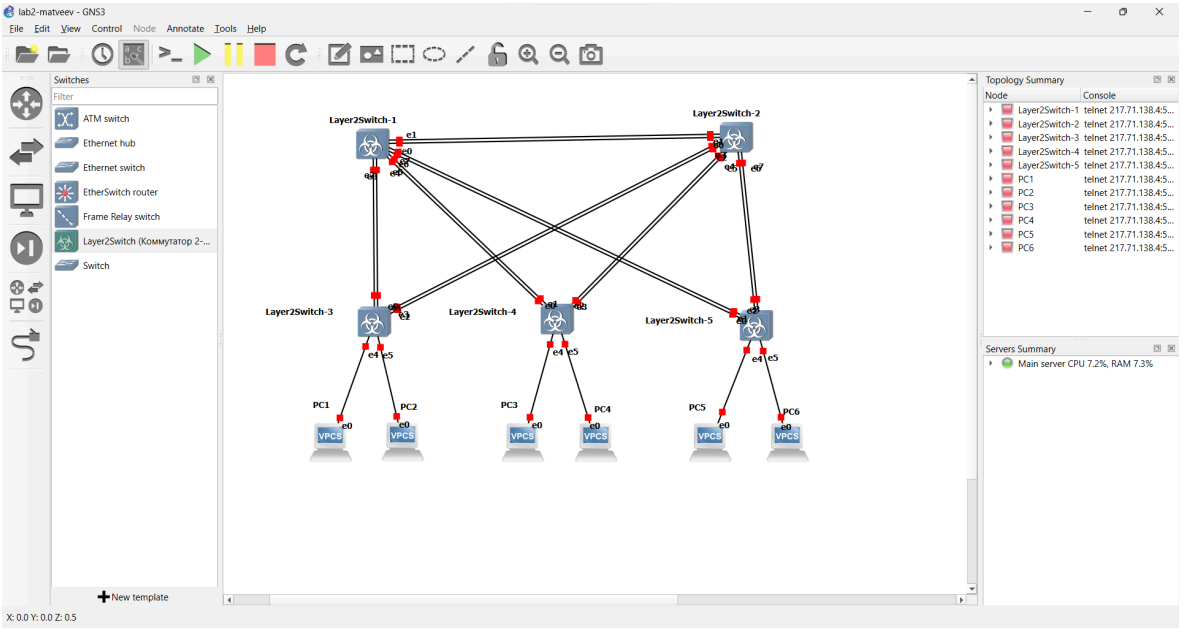
1) Для заданной на схеме schema-lab2 сети, состоящей из управляемых коммутаторов и персональных компьютеров настроить протокол STP, назначив явно один из коммутаторов корневым настройкой приоритета

Скопировал lab2-template проект.



Назначим корневым Layer2Switch-2

```
vIOS-L2-01>show vlan brief

VLAN Name                             Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                          active    Gi2/0
100  VLAN100                          active
200  VLAN0200                         active
300  VLAN0300                         active
1002 fddi-default                     act/unsup
1003 trcrf-default                    act/unsup
1004 fddinet-default                  act/unsup
1005 trbrf-default                    act/unsup

vIOS-L2-01>enable
vIOS-L2-01#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#
vIOS-L2-01(config)#spanning-tree vlan 1 priority 0
vIOS-L2-01(config)#exit

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0cf5.224d.0000
```

```
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    1      (priority 0 sys-id-ext 1)
             Address     0cf5.224d.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/0               Desg FWD 4         128.1    Shr
Gi0/1               Desg FWD 4         128.2    Shr
Gi0/2               Desg FWD 4         128.3    Shr
Gi0/3               Desg FWD 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
Gi1/2               Desg FWD 4         128.7    Shr
Gi1/3               Desg FWD 4         128.8    Shr

// Проверка со Layer2Switch-1
vIOS-L2-01>show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0cf5.224d.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0cff.2ed7.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/0               Root FWD 4         128.1    Shr
Gi0/1               Altn BLK 4         128.2    Shr
Gi0/2               Altn BLK 4         128.3    Shr
Gi0/3               Altn BLK 4         128.4    Shr
Gi1/0               Altn BLK 4         128.5    Shr
Gi1/1               Altn BLK 4         128.6    Shr
Gi1/2               Altn BLK 4         128.7    Shr
Gi1/3               Altn BLK 4         128.8    Shr
Gi2/0               Desg FWD 4         128.9    Shr
```

Для Layer2Switch-2 show spanning-tree выводит Root ID и Bridge ID одинаковые, также указано This bridge is the root.

2) Проверить доступность каждого с каждым всех персональных компьютеров (VPCS), результаты запротоколировать.

Установил для каждого VPCS IP адрес в одной сети (вида 192.168.0.x, где x - от 11 до 16).

```
PC1> ip 192.168.0.11/24
Checking for duplicate address...
PC1 : 192.168.0.11 255.255.255.0
PC1> save
Saving startup configuration to startup.vpc
.  done
```

Проверка со стороны PC1:

```
PC1> ping 192.168.0.12

84 bytes from 192.168.0.12 icmp_seq=1 ttl=64 time=7.720     ms
84 bytes from 192.168.0.12 icmp_seq=2 ttl=64 time=1.943     ms
^C
PC1> ping 192.168.0.13

84 bytes from 192.168.0.13 icmp_seq=1 ttl=64 time=8.419 ms
84 bytes from 192.168.0.13 icmp_seq=2 ttl=64 time=2.317 ms
^C
PC1> ping 192.168.0.14

84 bytes from 192.168.0.14 icmp_seq=1 ttl=64 time=9.929 ms
^C
PC1> ping 192.168.0.15

84 bytes from 192.168.0.15 icmp_seq=1 ttl=64 time=8.437 ms
84 bytes from 192.168.0.15 icmp_seq=2 ttl=64 time=22.341 ms
^C
PC1> ping 192.168.0.16

84 bytes from 192.168.0.16 icmp_seq=1 ttl=64 time=6.909 ms
^C
```

Проверка со стороны PC2:

```
PC2> ping 192.168.0.11

84 bytes from 192.168.0.11 icmp_seq=1 ttl=64 time=1.258 ms
^C
PC2> ping 192.168.0.13

84 bytes from 192.168.0.13 icmp_seq=1 ttl=64 time=8.563 ms
^C
PC2> ping 192.168.0.14
```

```
84 bytes from 192.168.0.14 icmp_seq=1 ttl=64 time=9.053 ms
^C
PC2> ping 192.168.0.15

84 bytes from 192.168.0.15 icmp_seq=1 ttl=64 time=5.369 ms
^C
PC2> ping 192.168.0.16

84 bytes from 192.168.0.16 icmp_seq=1 ttl=64 time=9.939 ms
^C
PC2> ping 192.168.0.16
```

Проверка со стороны PC3:

```
PC3> ping 192.168.0.11

84 bytes from 192.168.0.11 icmp_seq=1 ttl=64 time=7.239 ms
84 bytes from 192.168.0.11 icmp_seq=2 ttl=64 time=6.624 ms
84 bytes from 192.168.0.11 icmp_seq=3 ttl=64 time=3.923 ms
84 bytes from 192.168.0.11 icmp_seq=4 ttl=64 time=5.113 ms
^C
PC3> ping 192.168.0.12

84 bytes from 192.168.0.12 icmp_seq=1 ttl=64 time=7.463 ms
84 bytes from 192.168.0.12 icmp_seq=2 ttl=64 time=5.873 ms
84 bytes from 192.168.0.12 icmp_seq=3 ttl=64 time=7.304 ms
84 bytes from 192.168.0.12 icmp_seq=4 ttl=64 time=6.426 ms
84 bytes from 192.168.0.12 icmp_seq=5 ttl=64 time=8.911 ms

PC3> ping 192.168.0.14

84 bytes from 192.168.0.14 icmp_seq=1 ttl=64 time=4.335 ms
84 bytes from 192.168.0.14 icmp_seq=2 ttl=64 time=6.530 ms
84 bytes from 192.168.0.14 icmp_seq=3 ttl=64 time=0.935 ms
84 bytes from 192.168.0.14 icmp_seq=4 ttl=64 time=1.611 ms
84 bytes from 192.168.0.14 icmp_seq=5 ttl=64 time=0.901 ms

PC3> ping 192.168.0.15

84 bytes from 192.168.0.15 icmp_seq=1 ttl=64 time=15.743 ms
84 bytes from 192.168.0.15 icmp_seq=2 ttl=64 time=3.029 ms
84 bytes from 192.168.0.15 icmp_seq=3 ttl=64 time=7.847 ms
84 bytes from 192.168.0.15 icmp_seq=4 ttl=64 time=7.344 ms
84 bytes from 192.168.0.15 icmp_seq=5 ttl=64 time=3.925 ms

PC3> ping 192.168.0.16

84 bytes from 192.168.0.16 icmp_seq=1 ttl=64 time=2.037 ms
84 bytes from 192.168.0.16 icmp_seq=2 ttl=64 time=9.778 ms
84 bytes from 192.168.0.16 icmp_seq=3 ttl=64 time=1.969 ms
84 bytes from 192.168.0.16 icmp_seq=4 ttl=64 time=16.886 ms
84 bytes from 192.168.0.16 icmp_seq=5 ttl=64 time=6.882 ms
```

Проверка со стороны PC4:

```
PC4> ping 192.168.0.11

84 bytes from 192.168.0.11 icmp_seq=1 ttl=64 time=2.426 ms
84 bytes from 192.168.0.11 icmp_seq=2 ttl=64 time=6.882 ms
84 bytes from 192.168.0.11 icmp_seq=3 ttl=64 time=7.949 ms
84 bytes from 192.168.0.11 icmp_seq=4 ttl=64 time=8.349 ms
84 bytes from 192.168.0.11 icmp_seq=5 ttl=64 time=11.682 ms

PC4> ping 192.168.0.12

84 bytes from 192.168.0.12 icmp_seq=1 ttl=64 time=3.804 ms
84 bytes from 192.168.0.12 icmp_seq=2 ttl=64 time=5.550 ms
84 bytes from 192.168.0.12 icmp_seq=3 ttl=64 time=2.068 ms
84 bytes from 192.168.0.12 icmp_seq=4 ttl=64 time=8.209 ms
84 bytes from 192.168.0.12 icmp_seq=5 ttl=64 time=5.629 ms

PC4> ping 192.168.0.13

84 bytes from 192.168.0.13 icmp_seq=1 ttl=64 time=8.521 ms
84 bytes from 192.168.0.13 icmp_seq=2 ttl=64 time=2.681 ms
84 bytes from 192.168.0.13 icmp_seq=3 ttl=64 time=0.624 ms
84 bytes from 192.168.0.13 icmp_seq=4 ttl=64 time=3.112 ms
84 bytes from 192.168.0.13 icmp_seq=5 ttl=64 time=7.635 ms

PC4> ping 192.168.0.15

84 bytes from 192.168.0.15 icmp_seq=1 ttl=64 time=8.171 ms
84 bytes from 192.168.0.15 icmp_seq=2 ttl=64 time=6.356 ms
84 bytes from 192.168.0.15 icmp_seq=3 ttl=64 time=7.809 ms
84 bytes from 192.168.0.15 icmp_seq=4 ttl=64 time=7.477 ms
84 bytes from 192.168.0.15 icmp_seq=5 ttl=64 time=8.030 ms

PC4> ping 192.168.0.16

84 bytes from 192.168.0.16 icmp_seq=1 ttl=64 time=10.763 ms
84 bytes from 192.168.0.16 icmp_seq=2 ttl=64 time=5.643 ms
84 bytes from 192.168.0.16 icmp_seq=3 ttl=64 time=8.556 ms
84 bytes from 192.168.0.16 icmp_seq=4 ttl=64 time=2.039 ms
84 bytes from 192.168.0.16 icmp_seq=5 ttl=64 time=13.060 ms
```

Проверка со стороны PC5:

```
PC5> ping 192.168.0.11

84 bytes from 192.168.0.11 icmp_seq=1 ttl=64 time=5.285 ms
84 bytes from 192.168.0.11 icmp_seq=2 ttl=64 time=5.917 ms
84 bytes from 192.168.0.11 icmp_seq=3 ttl=64 time=16.169 ms
84 bytes from 192.168.0.11 icmp_seq=4 ttl=64 time=10.254 ms
84 bytes from 192.168.0.11 icmp_seq=5 ttl=64 time=10.827 ms

PC5> ping 192.168.0.12
```

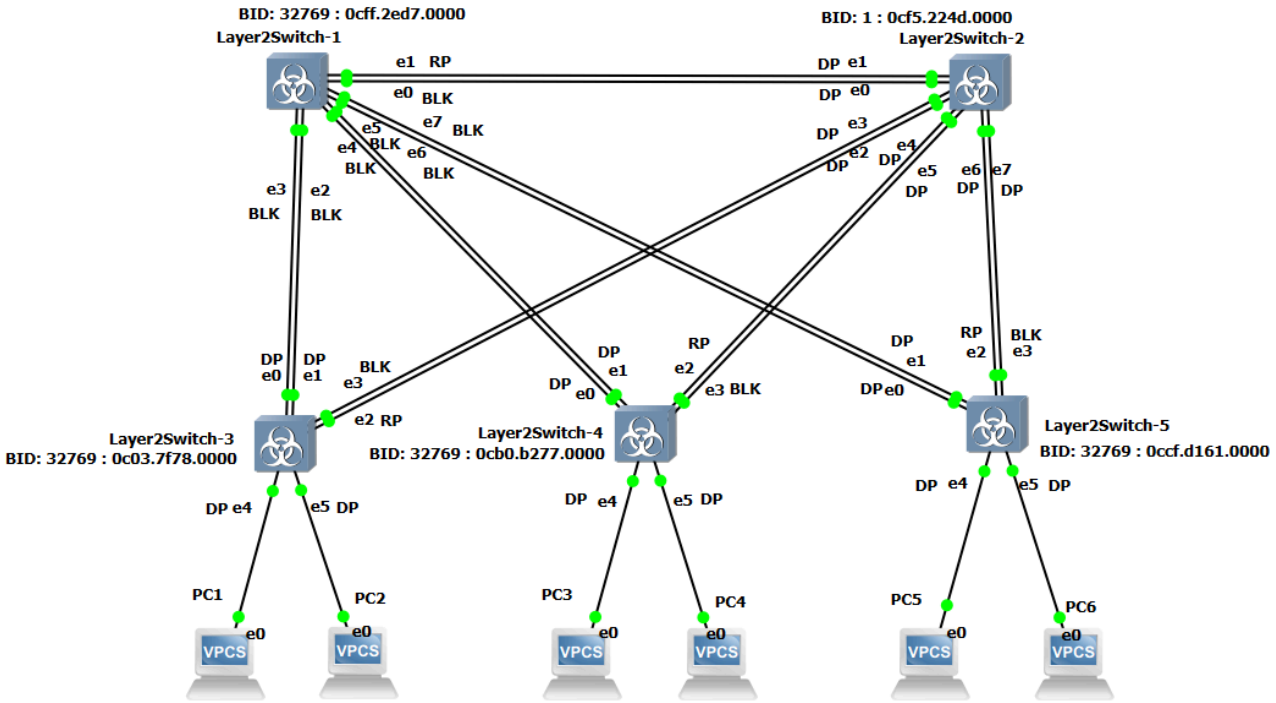
```
84 bytes from 192.168.0.12 icmp_seq=1 ttl=64 time=2.003 ms
84 bytes from 192.168.0.12 icmp_seq=2 ttl=64 time=1.908 ms
84 bytes from 192.168.0.12 icmp_seq=3 ttl=64 time=9.540 ms
84 bytes from 192.168.0.12 icmp_seq=4 ttl=64 time=2.700 ms
84 bytes from 192.168.0.12 icmp_seq=5 ttl=64 time=10.046 ms

PC5> ping 192.168.0.13

84 bytes from 192.168.0.13 icmp_seq=1 ttl=64 time=6.022 ms
84 bytes from 192.168.0.13 icmp_seq=2 ttl=64 time=13.464 ms
84 bytes from 192.168.0.13 icmp_seq=3 ttl=64 time=3.383 ms
84 bytes from 192.168.0.13 icmp_seq=4 ttl=64 time=10.804 ms
84 bytes from 192.168.0.13 icmp_seq=5 ttl=64 time=6.925 ms

PC5> ping 192.168.0.14

84 bytes from 192.168.0.14 icmp_seq=1 ttl=64 time=7.932 ms
84 bytes from 192.168.0.14 icmp_seq=2 ttl=64 time=7.469 ms
84 bytes from 192.168.0.14 icmp_seq=3 ttl=64 time=3.950 ms
84 bytes from 192.168.0.14 icmp_seq=4 ttl=64 time=2.058 ms
84 bytes from 192.168.0.14 icmp_seq=5 ttl=64 time=7.779 ms

PC5> ping 192.168.0.16

84 bytes from 192.168.0.16 icmp_seq=1 ttl=64 time=2.064 ms
84 bytes from 192.168.0.16 icmp_seq=2 ttl=64 time=2.687 ms
84 bytes from 192.168.0.16 icmp_seq=3 ttl=64 time=6.413 ms
84 bytes from 192.168.0.16 icmp_seq=4 ttl=64 time=8.470 ms
84 bytes from 192.168.0.16 icmp_seq=5 ttl=64 time=1.839 ms
```

Проверка со стороны PC6:

```
PC6> ping 192.168.0.11

84 bytes from 192.168.0.11 icmp_seq=1 ttl=64 time=7.275 ms
84 bytes from 192.168.0.11 icmp_seq=2 ttl=64 time=7.944 ms
84 bytes from 192.168.0.11 icmp_seq=3 ttl=64 time=6.250 ms
84 bytes from 192.168.0.11 icmp_seq=4 ttl=64 time=22.736 ms
84 bytes from 192.168.0.11 icmp_seq=5 ttl=64 time=9.507 ms

PC6> ping 192.168.0.12

84 bytes from 192.168.0.12 icmp_seq=1 ttl=64 time=8.625 ms
84 bytes from 192.168.0.12 icmp_seq=2 ttl=64 time=13.147 ms
84 bytes from 192.168.0.12 icmp_seq=3 ttl=64 time=7.121 ms
84 bytes from 192.168.0.12 icmp_seq=4 ttl=64 time=23.690 ms
84 bytes from 192.168.0.12 icmp_seq=5 ttl=64 time=4.863 ms

PC6> ping 192.168.0.13

84 bytes from 192.168.0.13 icmp_seq=1 ttl=64 time=5.342 ms
84 bytes from 192.168.0.13 icmp_seq=2 ttl=64 time=7.697 ms
84 bytes from 192.168.0.13 icmp_seq=3 ttl=64 time=6.066 ms
84 bytes from 192.168.0.13 icmp_seq=4 ttl=64 time=5.217 ms
```

```
84 bytes from 192.168.0.13 icmp_seq=5 ttl=64 time=9.923 ms

PC6> ping 192.168.0.14

84 bytes from 192.168.0.14 icmp_seq=1 ttl=64 time=7.947 ms
84 bytes from 192.168.0.14 icmp_seq=2 ttl=64 time=3.344 ms
84 bytes from 192.168.0.14 icmp_seq=3 ttl=64 time=9.070 ms
84 bytes from 192.168.0.14 icmp_seq=4 ttl=64 time=7.323 ms
84 bytes from 192.168.0.14 icmp_seq=5 ttl=64 time=3.408 ms

PC6> ping 192.168.0.15

84 bytes from 192.168.0.15 icmp_seq=1 ttl=64 time=3.523 ms
84 bytes from 192.168.0.15 icmp_seq=2 ttl=64 time=6.033 ms
84 bytes from 192.168.0.15 icmp_seq=3 ttl=64 time=6.490 ms
84 bytes from 192.168.0.15 icmp_seq=4 ttl=64 time=7.941 ms
84 bytes from 192.168.0.15 icmp_seq=5 ttl=64 time=8.991 ms
```

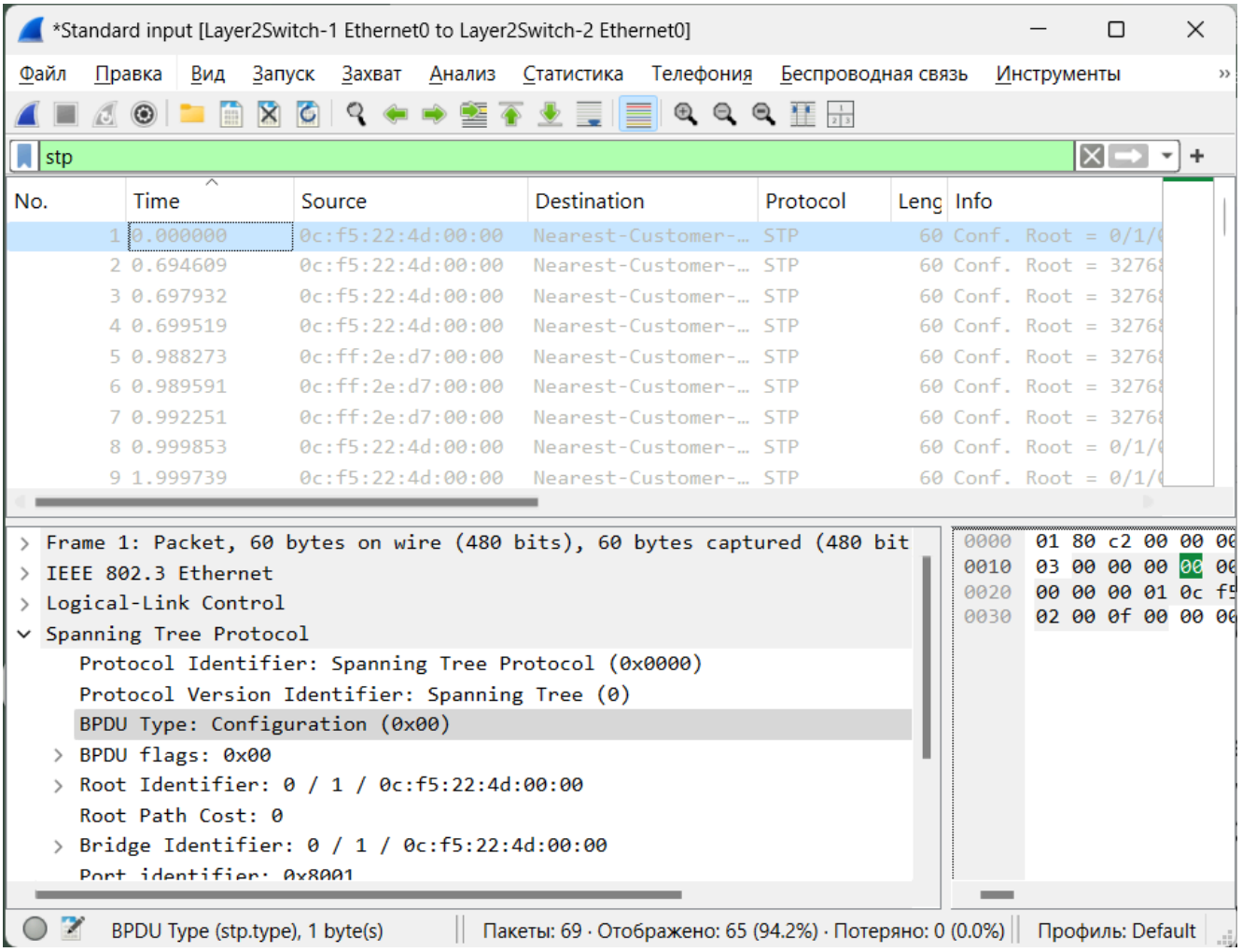
3) На изображении схемы отметить BID каждого коммутатора и режимы работы портов (RP/DP/blocked) и стоимости маршрутов, результат сохранить в файл

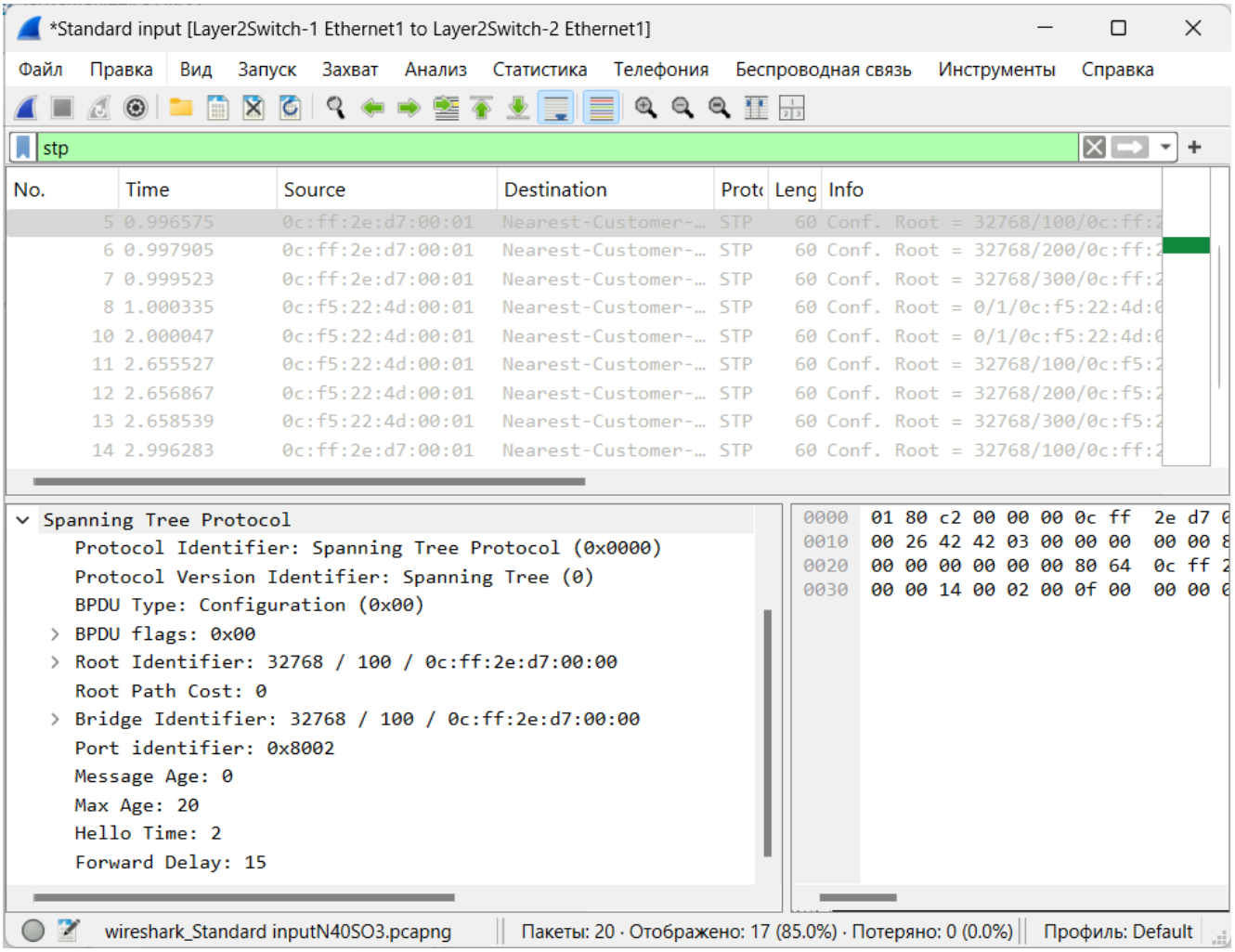


Информация о состоянии портов на каждом коммутаторе была получена с помощью команды show spanning-tree.

Из схемы следует, что для избавления от цикла в графе, Layer2Switch-1 заблокировал все свои порты.

4) При помощи wireshark отследить передачу пакетов hello от корневого коммутатора на всех линках (nb!), результаты включить в отчет




*Standard input [Layer2Switch-1 Ethernet1 to Layer2Switch-2 Ethernet1]
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка
stp
No. Time Source Destination Proto Leng Info
5 0.996575 0c:ff:2e:d7:00:01 Nearest-Customer-… STP 60 Conf. Root = 32768/100/0c:ff:2
6 0.997905 0c:ff:2e:d7:00:01 Nearest-Customer-… STP 60 Conf. Root = 32768/200/0c:ff:2
7 0.999523 0c:ff:2e:d7:00:01 Nearest-Customer-… STP 60 Conf. Root = 32768/300/0c:ff:2
8 1.000335 0c:f5:22:4d:00:01 Nearest-Customer-… STP 60 Conf. Root = 0/1/0c:f5:22:4d:0
10 2.000047 0c:f5:22:4d:00:01 Nearest-Customer-… STP 60 Conf. Root = 0/1/0c:f5:22:4d:0
11 2.655527 0c:f5:22:4d:00:01 Nearest-Customer-… STP 60 Conf. Root = 32768/100/0c:f5:2
12 2.656867 0c:f5:22:4d:00:01 Nearest-Customer-… STP 60 Conf. Root = 32768/200/0c:f5:2
13 2.658539 0c:f5:22:4d:00:01 Nearest-Customer-… STP 60 Conf. Root = 32768/300/0c:f5:2
14 2.996283 0c:ff:2e:d7:00:01 Nearest-Customer-… STP 60 Conf. Root = 32768/100/0c:ff:2
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 32768 / 100 / 0c:ff:2e:d7:00:00
Root Path Cost: 0
Bridge Identifier: 32768 / 100 / 0c:ff:2e:d7:00:00
Port identifier: 0x8002
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15
0000 01 80 c2 00 00 00 0c ff 2e d7
0010 00 26 42 42 03 00 00 00 00 00
0020 00 00 00 00 00 00 80 64 0c ff
0030 00 00 14 00 02 00 0f 00 00 00
wireshark_Standard inputN40SO3.pcapng
Пакеты: 20 · Отображено: 17 (85.0%) · Потеряно: 0 (0.0%)
Профиль: Default

*Standard input [Layer2Switch-3 Ethernet3 to Layer2Switch-2 Ethernet3]

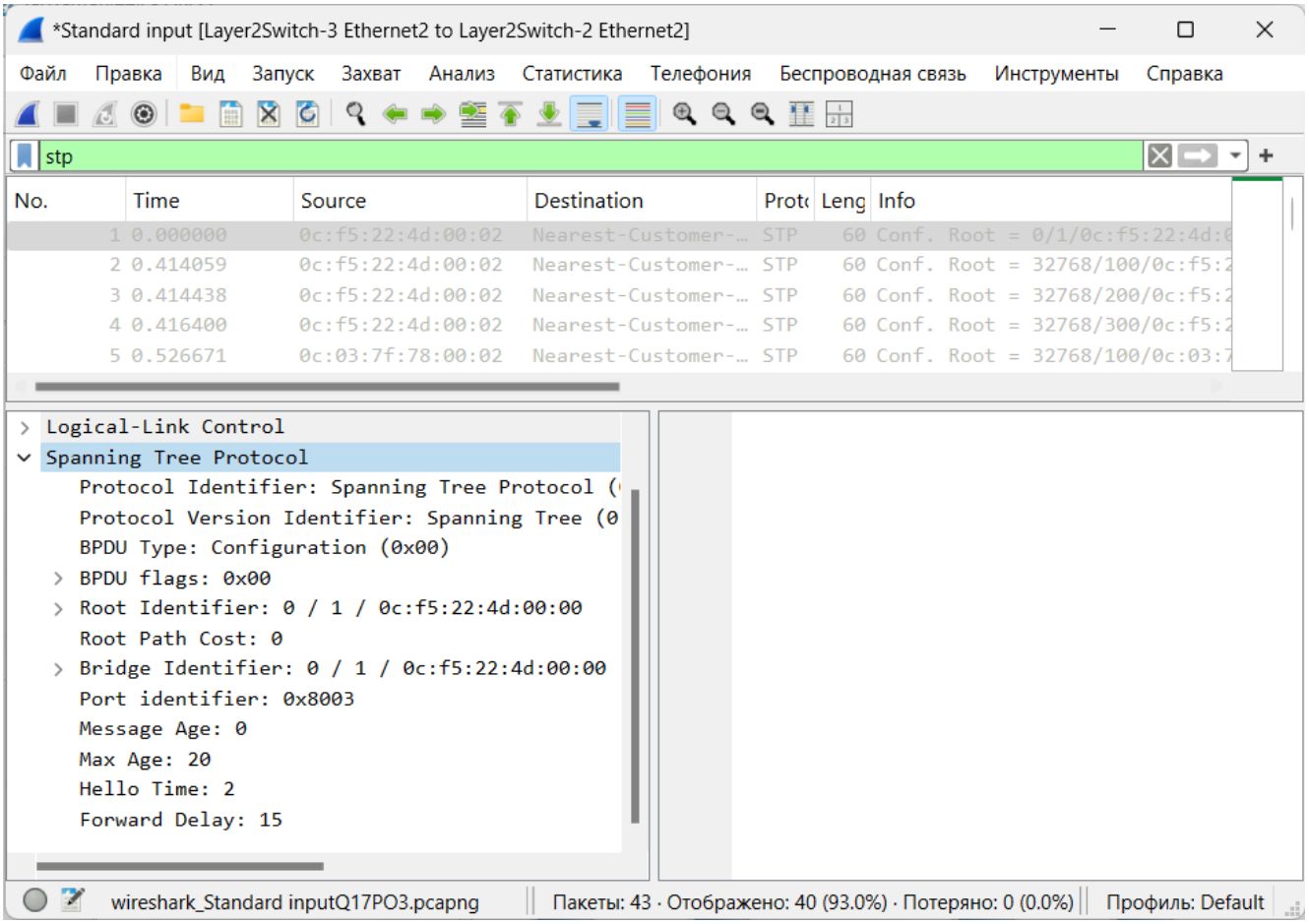
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp

No.	Time	Source	Destination	Proto	Length	Info
1	0.000000	0c:03:7f:78:00:03	Nearest-Customer-…	STP	60	Conf. Root = 32768/100/0c:03
2	0.000121	0c:03:7f:78:00:03	Nearest-Customer-…	STP	60	Conf. Root = 32768/200/0c:03
3	0.001904	0c:03:7f:78:00:03	Nearest-Customer-…	STP	60	Conf. Root = 32768/300/0c:03
4	0.462175	0c:f5:22:4d:00:03	Nearest-Customer-…	STP	60	Conf. Root = 0/1/0c:f5:22:4d
5	1.462012	0c:f5:22:4d:00:03	Nearest-Customer-…	STP	60	Conf. Root = 0/1/0c:f5:22:4d
6	1.874381	0c:f5:22:4d:00:03	Nearest-Customer-…	STP	60	Conf. Root = 32768/100/0c:f5
7	1.875388	0c:f5:22:4d:00:03	Nearest-Customer-…	STP	60	Conf. Root = 32768/200/0c:f5
8	1.877043	0c:f5:22:4d:00:03	Nearest-Customer-…	STP	60	Conf. Root = 32768/300/0c:f5
9	1.999604	0c:03:7f:78:00:03	Nearest-Customer-…	STP	60	Conf. Root = 32768/100/0c:03
10	1.999705	0c:03:7f:78:00:03	Nearest-Customer-…	STP	60	Conf. Root = 32768/200/0c:03
11	2.001573	0c:03:7f:78:00:03	Nearest-Customer-…	STP	60	Conf. Root = 32768/300/0c:03

```
> Ethernet II, Src: 0c:03:7f:78:00:03 (0c:03:7f:78:00:03), Dst: Nearest-
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 100
> Logical-Link Control
v Spanning Tree Protocol
    Protocol Identifier: Spanning Tree Protocol (0x0000)
    Protocol Version Identifier: Spanning Tree (0)
    BPDU Type: Configuration (0x00)
  > BPDU flags: 0x00
  > Root Identifier: 32768 / 100 / 0c:03:7f:78:00:00
    Root Path Cost: 0
  > Bridge Identifier: 32768 / 100 / 0c:03:7f:78:00:00
    Port identifier: 0x8004
    Message Age: 0
    Max Age: 20
    Hello Time: 2
    Forward Delay: 15
```

wireshark_Standard inputP4LJO3.pcapng | Пакеты: 63 · Отображено: 59 (93.7%) · Потеряно: 0 (0.0%) | Профиль: Default


*Standard input [Layer2Switch-3 Ethernet2 to Layer2Switch-2 Ethernet2]
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка
stp
No. Time Source Destination Prot Leng Info
1 0.000000 0c:f5:22:4d:00:02 Nearest-Customer-… STP 60 Conf. Root = 0/1/0c:f5:22:4d:0
2 0.414059 0c:f5:22:4d:00:02 Nearest-Customer-… STP 60 Conf. Root = 32768/100/0c:f5:2
3 0.414438 0c:f5:22:4d:00:02 Nearest-Customer-… STP 60 Conf. Root = 32768/200/0c:f5:2
4 0.416400 0c:f5:22:4d:00:02 Nearest-Customer-… STP 60 Conf. Root = 32768/300/0c:f5:2
5 0.526671 0c:03:7f:78:00:02 Nearest-Customer-… STP 60 Conf. Root = 32768/100/0c:03:7
Logical-Link Control
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (
Protocol Version Identifier: Spanning Tree (0
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 0 / 1 / 0c:f5:22:4d:00:00
Root Path Cost: 0
Bridge Identifier: 0 / 1 / 0c:f5:22:4d:00:00
Port identifier: 0x8003
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15
wireshark_Standard inputQ17PO3.pcapng
Пакеты: 43 · Отображено: 40 (93.0%) · Потеряно: 0 (0.0%)
Профиль: Default

*Standard input [Layer2Switch-2 Ethernet4 to Layer2Switch-4 Ethernet2]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты

stp

No.	Time	Source	Destination	Protocol	Leng	Info
1	0.000000	0c:b0:b2:77:00:02	Nearest-Customer-…	STP	60	Conf. Root = 32768
2	0.001720	0c:b0:b2:77:00:02	Nearest-Customer-…	STP	60	Conf. Root = 32768
3	0.002975	0c:b0:b2:77:00:02	Nearest-Customer-…	STP	60	Conf. Root = 32768

```
> Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits)
> Ethernet II, Src: 0c:b0:b2:77:00:02 (0c:b0:b2:77:00:02), Dst: Nearest-Custo
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 100
> Logical-Link Control
v Spanning Tree Protocol
    Protocol Identifier: Spanning Tree Protocol (0x0000)
    Protocol Version Identifier: Spanning Tree (0)
    BPDU Type: Configuration (0x00)
  > BPDU flags: 0x00
  > Root Identifier: 32768 / 100 / 0c:b0:b2:77:00:00
    Root Path Cost: 0
  > Bridge Identifier: 32768 / 100 / 0c:b0:b2:77:00:00
    Port identifier: 0x8003
    Message Age: 0
    Max Age: 20
    Hello Time: 2
    Forward Delay: 15
```

Spanning Tree Protocol (stp), 35 byte(s) | Пакеты: 26 · Отображено: 24 (92.3%) · Потеряно: 0 (0.0%) | Профиль: Default

Захват с Standard input [Layer2Switch-2 Ethernet5 to Layer2Switch-4 Ethernet3]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты

stp

No.	Time	Source	Destination	Protocol	Leng	Info
37	8.000753	0c:f5:22:4d:00:05	Nearest-Customer-…	STP	60	Conf. Root = 32768/300
38	8.003528	0c:f5:22:4d:00:05	Nearest-Customer-…	STP	60	Conf. Root = 0/1/0c:f5

```
> Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits)
> Ethernet II, Src: 0c:f5:22:4d:00:05 (0c:f5:22:4d:00:05), Dst: Nearest-Custo
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 100
> Logical-Link Control
v Spanning Tree Protocol
    Protocol Identifier: Spanning Tree Protocol (0x0000)
    Protocol Version Identifier: Spanning Tree (0)
    BPDU Type: Configuration (0x00)
  > BPDU flags: 0x00
  > Root Identifier: 32768 / 100 / 0c:f5:22:4d:00:00
    Root Path Cost: 0
  > Bridge Identifier: 32768 / 100 / 0c:f5:22:4d:00:00
    Port identifier: 0x8006
    Message Age: 0
    Max Age: 20
    Hello Time: 2
    Forward Delay: 15
```

```
0000  01 80 c2 00 00 00
0010  00 26 42 42 03 00
0020  00 00 00 00 00 00
0030  00 00 14 00 02 00
```

Spanning Tree Protocol: Protocol | Пакеты: 38 · Отображено: 36 (94.7%) | Профиль: Default

*Standard input [Layer2Switch-2 Ethernet6 to Layer2Switch-5 Ethernet2]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты

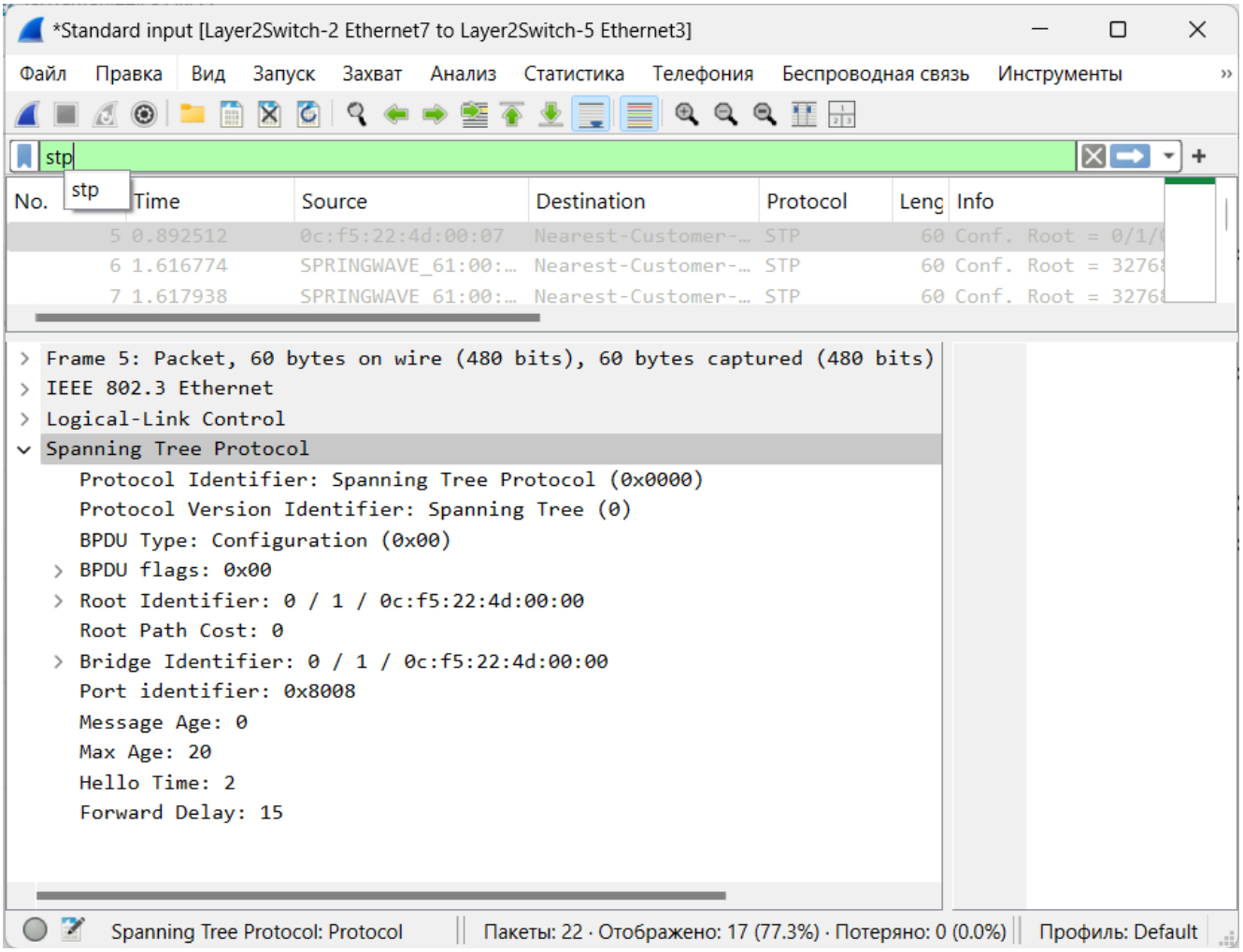
stp

No.	Time	Source	Destination	Protocol	Leng	Info
1	0.000000	SPRINGWAVE_61:00:…	Nearest-Customer-…	STP	60	Conf. Root = 3276
2	0.001133	SPRINGWAVE_61:00:…	Nearest-Customer-…	STP	60	Conf. Root = 3276
3	0.002141	SPRINGWAVE_61:00:…	Nearest-Customer-…	STP	60	Conf. Root = 3276

```
> Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits)
> Ethernet II, Src: SPRINGWAVE_61:00:02 (0c:cf:d1:61:00:02), Dst: Nearest-Cus
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 100
> Logical-Link Control
v Spanning Tree Protocol
    Protocol Identifier: Spanning Tree Protocol (0x0000)
    Protocol Version Identifier: Spanning Tree (0)
    BPDU Type: Configuration (0x00)
  > BPDU flags: 0x00
  > Root Identifier: 32768 / 100 / 0c:cf:d1:61:00:00
    Root Path Cost: 0
  > Bridge Identifier: 32768 / 100 / 0c:cf:d1:61:00:00
    Port identifier: 0x8003
    Message Age: 0
    Max Age: 20
    Hello Time: 2
    Forward Delay: 15
```

```
0000  01 80 c2 00 00 0
0010  00 26 42 42 03 0
0020  00 00 00 00 00 0
0030  00 00 14 00 02 0
```

wireshark_Standa...put5QL0O3.pcapng | Пакеты: 32 · Отображено: 31 (96.9%) · Потеряно: 0 (0.0%) | Профиль: Default



5) Изменить стоимость маршрута для порта RP произвольного назначенного (designated) коммутатора, повторить действия из п.3, результат сохранить в отдельный файл

Для коммутатора-1 смотрим, Gi0/0 (Ethernet0) это Root port, подключен к корню.

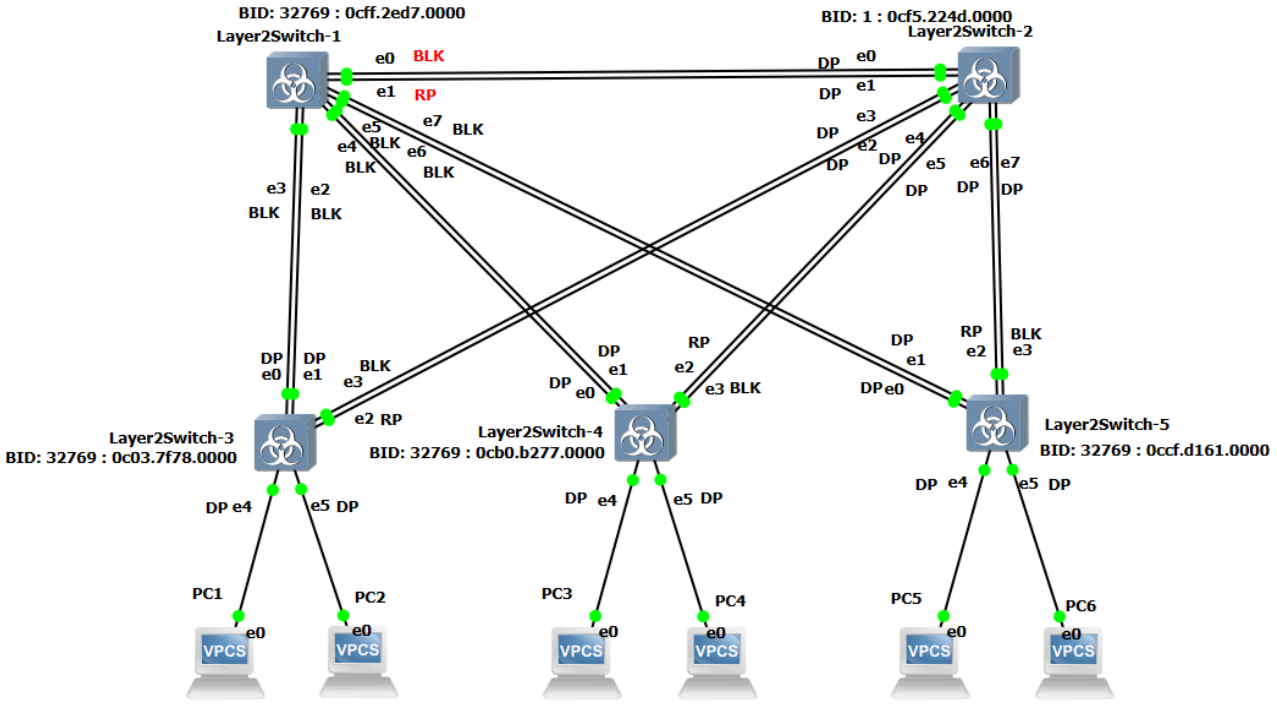
```
vIOS-L2-01>show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0cf5.224d.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0cff.2ed7.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/0               Root FWD 4         128.1    Shr
```

```
Gi0/1               Altn BLK 4         128.2    Shr
Gi0/2               Altn BLK 4         128.3    Shr
Gi0/3               Altn BLK 4         128.4    Shr
Gi1/0               Altn BLK 4         128.5    Shr
Gi1/1               Altn BLK 4         128.6    Shr
Gi1/2               Altn BLK 4         128.7    Shr
Gi1/3               Altn BLK 4         128.8    Shr
Gi2/0               Desg FWD 4         128.9    Shr
```

Изменим значение стоимости для gigabitEthernet0/0,

```
vIOS-L2-01(config)#interface gigabitEthernet0/1
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#interface gigabitEthernet0/0
vIOS-L2-01(config-if)#spanning-tree vlan 1 cost 999
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#exit


vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0cf5.224d.0000
             Cost        4
             Port        2 (GigabitEthernet0/1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0cff.2ed7.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/0               Altn BLK 999       128.1    Shr
Gi0/1               Root FWD 4         128.2    Shr
Gi0/2               Altn BLK 4         128.3    Shr
Gi0/3               Altn BLK 4         128.4    Shr
Gi1/0               Altn BLK 4         128.5    Shr
Gi1/1               Altn BLK 4         128.6    Shr
Gi1/2               Altn BLK 4         128.7    Shr
Gi1/3               Altn BLK 4         128.8    Shr
Gi2/0               Desg FWD 4         128.9    Shr
```

В выводе show spanning-tree видно, что с теперь Gi0/0 в режиме BLK, а режим Root теперь лежит на Gi0/1, который прежде был заблокированным.



6) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств